INMO(29^{th}) – 2014

Time – 4 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.

1. In a triangle ABC, let D be a point on the segment BC such that AB + BD = AC + CD. Suppose that the points B, C and the centroids of triangles ABD and ACD lie on a circle. Prove that AB = AC.
2. Let n be a natural number. Prove that $\left[\frac{n}{1}\right]+\left[\frac{n}{2}\right]+\left[\frac{n}{3}\right]+...+\left[\frac{n}{n}\right]+\left[\sqrt{n}\right]$ is even. (Here $[x]$ denotes the largest integer smaller than or equal to x).
3. Let a, b be natural numbers with $ab > 2$. Suppose that the sum of their greatest common divisor and least common multiple is divisible by a + b. Prove that the quotient is at most (a + b)/4. When is this quotient exactly equal to (a + b)/4?
4. Written on a blackboard is the polynomial $x^2 + x + 2014$. Calvin and Hobbes take turns alternatively (starting with Calvin) in the following game. During his turn, Calvin should either increase or decrease the coefficients of x by 1. And during his turn, Hobbes should either increase or decrease the constant the coefficient by 1. Calvin wins if at any point of time the polynomial on the blackboard at that instant has integer roots. Prove that Calvin has a winning strategy.
5. In an acute angled triangle ABC, a point D lies on the segment BC. Let O_1, O_2 denote the circumcentres of the triangles ABD and ACD respectively. Prove that the line joining the circumcentre of triangle ABC and the orthocenter of triangle O_1O_2D is parallel to BC.
6. Let n be a natural number and X = {1, 2, …, n}. For subsets A and B of X we define $A\Delta B$ to the set of all those elements of X which belongs to exactly one of A and B. Let F be a collection of subsets of X such that for any two distinct elements A and B in F the set $A\Delta B$ has at least two elements. Show that F has at most 2^{n-1} elements. Find all such collections F with 2^{n-1} elements.

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